Ex. No: 01

Image Handling and Pixel Transformations Using OpenCV

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```
# Import libraries
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

1. Read the image (‘Eagle_in_Flight.jpg’) using OpenCV imread() as a grayscale image.

```
img = cv2.imread("Eagle_in_Flight.jpg", cv2.IMREAD_GRAYSCALE)
```

2. Print the image width, height & Channel

```
img.shape
```

```
(600, 768)
```

3. Display the image using matplotlib imshow().

```
plt.imshow(img,cmap='gray')
plt.title("Gray Scale Image")
plt.axis("OFF")
plt.show()
```



4. Save the image as a PNG file using OpenCV imwrite().

```
cv2.imwrite("img.png", img)
```

```
True
```

5. Read the saved image above as a color image using cv2.cvtColor().

```
img1=cv2.imread("Eagle_in_Flight.jpg")
C_image=cv2.cvtColor(img1,cv2.COLOR_BGR2RGB)
plt.imshow(C_image,cmap='gray')
```

```
<matplotlib.image.AxesImage at 0x19b21342c10>
```



6. Display the Colour image using matplotlib imshow() & Print the image width, height & channel

```
C_image.shape
```

```
(600, 768, 3)
```

7. Crop the image to extract any specific (Eagle alone) object from the image.

```
eagle = C_image[20:410, 200:550]
plt.imshow(eagle)
plt.axis('off')
plt.show()
```



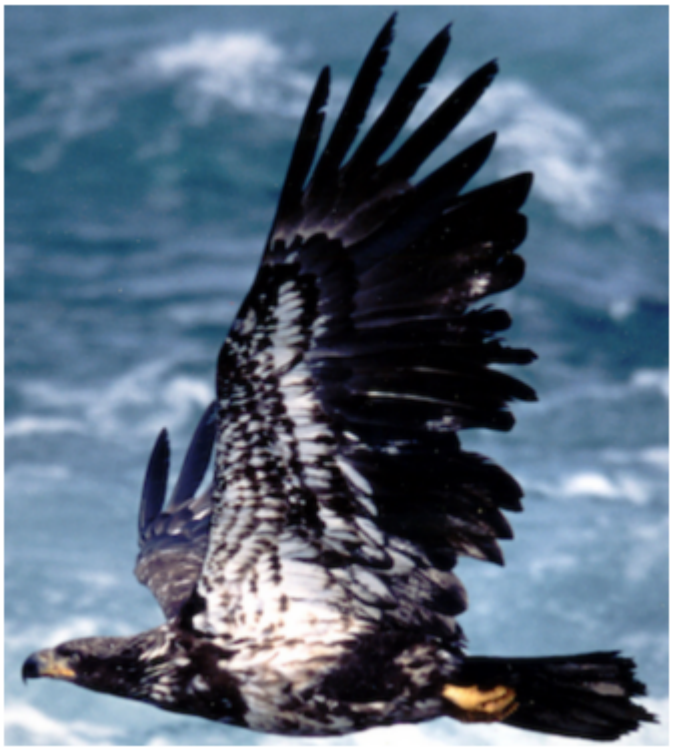
8. Resize the image up by a factor of 2x.

```
eagle_2x = cv2.resize(eagle, None, fx=2, fy=2, interpolation=cv2.INTER_LINEAR)
plt.imshow(eagle_2x)
plt.axis('off')
plt.show()
```



9. Flip the cropped/resized image horizontally. (Eagle Image)

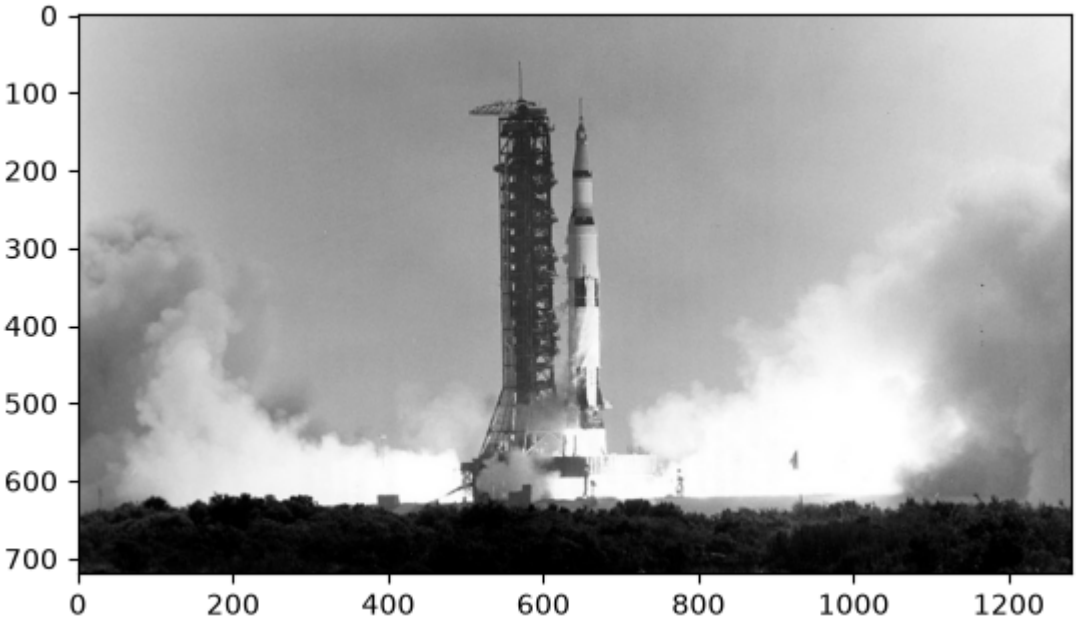
```
eagle_flip = cv2.flip(eagle_2x, 1)
plt.imshow(eagle_flip)
plt.axis('off')
plt.show()
```



10. Read in the image (‘Apollo-11-launch.jpg’).

```
img = cv2.imread('Apollo-11-launch.jpg')
plt.imshow(img[:, :, ::-1])
plt.show
```

```
<function matplotlib.pyplot.show(close=None, block=None)>
```



11. Add the following text to the dark area at the bottom of the image (centered on the image):

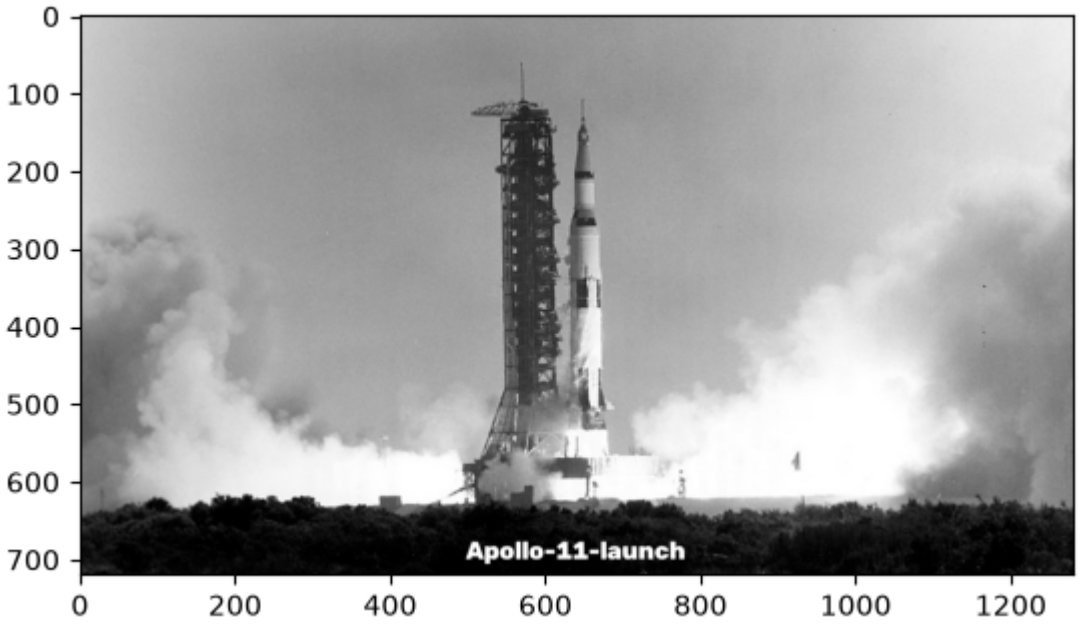
```
text = 'Apollo-11-launch'
font_face = cv2.FONT_HERSHEY_PLAIN
font_scale = 2
thickness = 2

(text_width, text_height), _ = cv2.getTextSize(text, font_face, font_scale, thickness)
```

```
x = (img.shape[1] - text_width) // 2
y = 700

cv2.putText(img, text, (x, y), font_face, font_scale, (255, 255, 255), thickness)

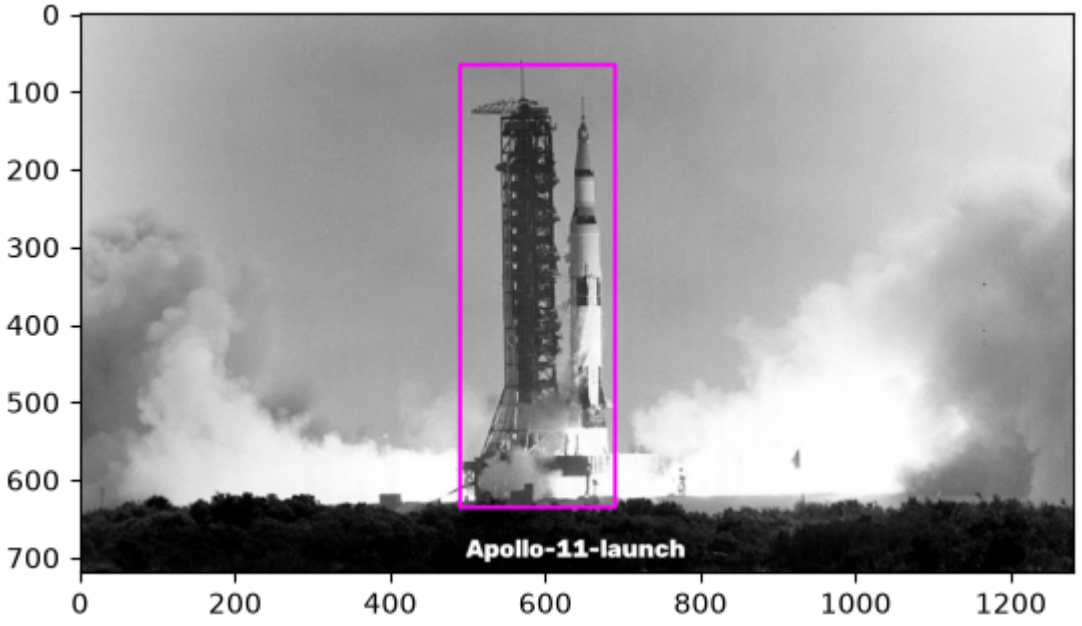
plt.imshow(img, cmap='gray')
plt.show()
```



12. Draw a magenta rectangle that encompasses the launch tower and the rocket.

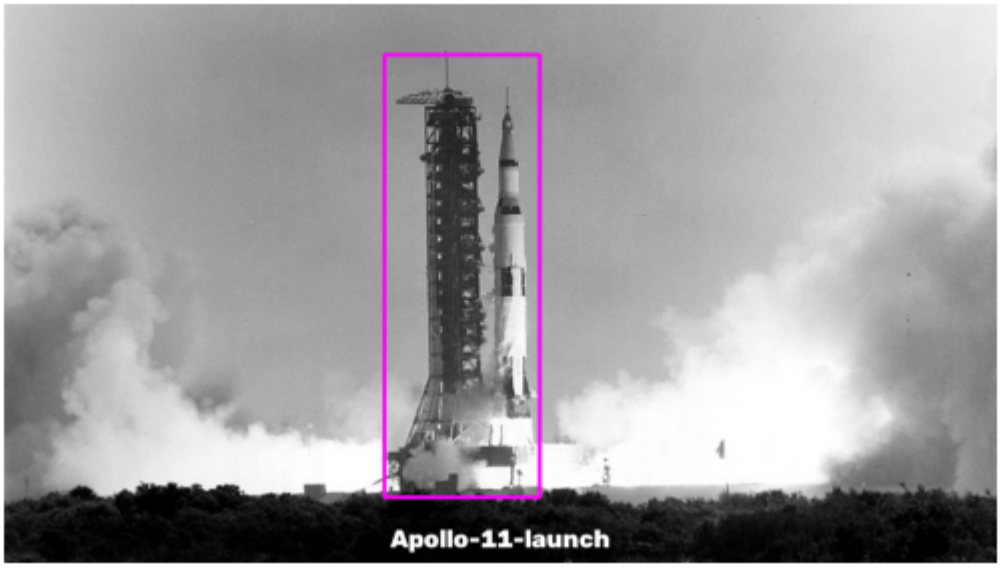
```
cv2.rectangle(img, (490, 65), (690, 635), (255, 0, 255), 3)

plt.imshow(img, cmap='gray')
plt.show()
```



13. Display the final annotated image.

```
plt.imshow(img, cmap='gray')
plt.axis('off')
plt.show()
```



14. Read the image ('boy.jpg').

```
img = cv2.imread('boy.jpg')
```

15. Adjust the brightness of the image.

```
brightness = 50
bright_img = cv2.convertScaleAbs(img, alpha=1, beta=brightness)

plt.imshow(cv2.cvtColor(bright_img, cv2.COLOR_BGR2RGB))
plt.axis('off')
plt.show()
```



16. Create brighter and darker images.

```
brighter = cv2.convertScaleAbs(img, alpha=1, beta=50)
darker = cv2.convertScaleAbs(img, alpha=1, beta=-50)
```

```
plt.imshow(cv2.cvtColor(brighter, cv2.COLOR_BGR2RGB))
plt.axis('off')
plt.show()

plt.imshow(cv2.cvtColor(darker, cv2.COLOR_BGR2RGB))
plt.axis('off')
plt.show()
```





17. Display the images (Original Image, Darker Image, Brighter Image).

```
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.axis('off')
plt.show()
```

```
plt.imshow(cv2.cvtColor(darker, cv2.COLOR_BGR2RGB))
plt.axis('off')
plt.show()

plt.imshow(cv2.cvtColor(brighter, cv2.COLOR_BGR2RGB))
plt.axis('off')
plt.show()
```







18. Modify the image contrast.

```
matrix1 = np.ones(img.shape) * 1.1
matrix2 = np.ones(img.shape) * 1.2
img_higher1 = cv2.multiply(img, 1.1)
img_higher2 = cv2.multiply(img, 1.2)
```

19. Display the images (Original, Lower Contrast, Higher Contrast).

```
lower_contrast = cv2.convertScaleAbs(img, alpha=0.5, beta=0)
higher_contrast = cv2.convertScaleAbs(img, alpha=1.5, beta=0)
plt.figure(figsize=(12, 4))
plt.subplot(1, 3, 1)
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title('Original Image')
plt.axis('off')
plt.subplot(1, 3, 2)
plt.imshow(cv2.cvtColor(lower_contrast, cv2.COLOR_BGR2RGB))
plt.title('Lower Contrast')
plt.axis('off')
plt.subplot(1, 3, 3)
plt.imshow(cv2.cvtColor(higher_contrast, cv2.COLOR_BGR2RGB))
plt.title('Higher Contrast')
plt.axis('off')
plt.show()
```



20. Split the image (boy.jpg) into the B,G,R components & Display the channels.

```
B, G, R = cv2.split(img)
plt.figure(figsize=(12, 4))
plt.subplot(1, 3, 1)
plt.imshow(B, cmap='gray')
plt.title('Blue Channel')
plt.axis('off')
plt.subplot(1, 3, 2)
plt.imshow(G, cmap='gray')
plt.title('Green Channel')
plt.axis('off')
plt.subplot(1, 3, 3)
plt.imshow(R, cmap='gray')
plt.title('Red Channel')
plt.axis('off')
plt.show()
```



21. _merged the R, G, B , display along with orginal image

```
merged = cv2.merge([B, G, R])
plt.figure(figsize=(8, 4))
plt.subplot(1, 2, 1)
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title('Original Image')
plt.axis('off')
plt.subplot(1, 2, 2)
plt.imshow(cv2.cvtColor(merged, cv2.COLOR_BGR2RGB))
plt.title('Merged Image')
plt.axis('off')
plt.show()
```



22. Split the image into the H, S, V components & Display the channels.

```
hsv = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)
H, S, V = cv2.split(hsv)
plt.figure(figsize=(12, 4))
plt.subplot(1, 3, 1)
plt.imshow(H, cmap='gray')
```

```
plt.title('Hue Channel')
plt.axis('off')
plt.subplot(1, 3, 2)
plt.imshow(S, cmap='gray')
plt.title('Saturation Channel')
plt.axis('off')
plt.subplot(1, 3, 3)
plt.imshow(V, cmap='gray')
plt.title('Value Channel')
plt.axis('off')
plt.show()
```



23. _merged the H, S, V, display along with orginal image

```
merged_hsv = cv2.merge([H, S, V])
plt.figure(figsize=(8, 4))
plt.subplot(1, 2, 1)
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title('Original Image')
plt.axis('off')
plt.subplot(1, 2, 2)
plt.imshow(cv2.cvtColor(merged_hsv, cv2.COLOR_HSV2RGB))
plt.title('Merged HSV Image')
plt.axis('off')
plt.show()
```

